11-711 Fall 2026 Homework 2

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Abstract

This document is a report for my homework 2. In this work, I developed an end-to-end RAG (Lewis et al., 2020) system to answer questions related to CMU and Pittsburgh. I have tried several retrieval, chunking, embedding, fusion, and LLM methods to achieve the best performance.

1 Data Creation

1. **How did you compile your knowledge resource, and how did you decide which documents to include?**

   Besides the sources from the handout, I have include some extra sources:
   1. School information: cmu.edu (Carnegie Mellon University), calender.pitt.edu (University of Pittsburgh), different college athletics website, and different university about pages
   2. Pittsburgh events: nextpittsburgh.com (NEXTpittsburgh)
   3. Sport's public api
   4. Millage and budget data from apps.alleghenycounty.us (Allegheny County)
   5. Museums: carnegie museum of art site (Carnegie Museum of Art), cmoa collection data (Carnegie Museum of Art, 2015)
   6. Stores in pittsburgh: openStreetMap's api (OpenStreetMap contributors, 2026)
   7. Libraries: ACLA site for libraries in Allegheny County (Allegheny County Library Association)

   Initially I start with the sources suggested by the handout. After evaluating the missed data from public evaluation questions, I found some information is not in the corpus I have, such as millage data and museum collection data. So I added these extra sites mentioned above. Instead of just adding what is needed for this question, I include the whole resources such as all milage, the full collection to make sure there will be enough information in the test time. Moreover, I add osm data to include every store's information in case in the test time, a random store in Pittsburgh is asked.

2. **How did you extract raw data? What tools did you use?**

   For Wikipedia pages, I use MediaWiki API (Wikimedia Foundation) so that the plain text returned is usable. And then, I turn it into markdown-style headings and drop the boilerplate, such as References, See also, etc. I keep the heading because it helps to preserve wiki structure so it can be used in chunking.

   For HTML, I use BeautifulSoup (Richardson) with lxml parser (Behnel et al.). It will use the og:title or <title> tag as title and delete unnecessary tags such as <nav>, <footer>, etc. It will then use the <main> part and preserve the overall structure by using markdown style. It will also remove repeated long lines (>=25 chars) within a page and remove lines on at least 30% of the site's pages. I also keep the social media information before the footer is removed, so questions like "What is the Arcade Comedy Theater's Instagram handle?" can be answered.

   For PDF (tax regulations, budgets), I use pdfplumber (Singer-Vine et al.) for my main extraction work. Since plain extraction cannot handle table format well, if there is table in the pdf, it will be added after the page text after a proper scan. And then, if the whole documents have <200 chars, it means that the pdfplumber failed. In this way, it will fall back to pypdf (py-pdf contributors). Any

pages with < 40 chars will go through OCR (pydium2 (pypdfium2 contributors) -> Tesseract (Tesseract OCR contributors)) because this pdf is likely scanned image. Finally it will skip short documents, delete duplications, and re-join hypen in a word ("infor-mation" -> "information") for increasing exact match scores.

3. **What data was annotated for training if any (what kind and how much)?**
   I did not annotate any data.

4. **For training data that you did not annotate, did you use any extra data and in what way?**
   I did not train or fine-tune models. The extra data I use is from both the handout and self-built. They are used as few-shot learning examples provided to the LLM reader model (Qwen 30B instruct (Team, 2025)).

2 Model details

2.1 System Variants

2.1.1 Answer Style

I have tried four answer styles: short, routed, full, restate. Different answer type comes from changing the prompt to the reader model. Short style tells model to keep answer under 10 words, routed style detects why/how/what questions with regex and give those specific question longer answer budget (<= 25 words), full style gives every questions longer answer budget, and restate style prompt model to restate the subject in the question with the answer in one sentence.

2.1.2 Reranker

I tried reranker to increase retrieval correctness. I use bge-reranker-v2-m3 (Chen et al., 2024) as my reranker. It is a cross-encoder that takes question and chunks as input and output a relevance score. The reranker rerank top-50 chunks and keep top 10. I have also tried Qwen3-Reranker-0.6/4B (Zhang et al., 2025) but the performace looks similar with extra compute.

2.2 Explanation

2.2.1 Answer style

Short style: the handout suggests to keep answers concise, because the metrics are token-based.
Full style: to keep the extra information that may be in the answer, like full names or a date together with the event.
Routed style: I notice that public evaluation data are mostly short, but descriptive type of questions have longer correct answer, so I route by question type.
Restate style: I found that in public evaluation data, longer answers are sentences that repeat the question's subject.

2.2.2 Reranker

The motivation behind using reranker is that I found some target chunks are missing from the top 10 list. In the hybrid approach I use, BM25 (Lù, 2024) computes word match and BGE (Xiao et al., 2023) compute chunk embedding and question embedding seperately. Both BM25 and BGE did not consider the semantic meaning of question and chunk together, so adding a cross-encoder reranker may be helpful. Also, for the reader LLM, it tends to memorizes contents at the start and at the end. Reranking will help to keep relevant information on top, improving reader model's performance.

3 Results

3.1 System Variants

Here I report two variants: reranker and answer style.

3.1.1 Reranker

In Table 1, I compared different reranker performances on different topk recall with different k (k = 1, 5, 10). The comparison is on 121 short answer questions from the public evaluation data. The reason for chosing short answer only is that I want to easily determine which chunk the answer is in. For shorter answers, this is easier and avoid potential error.

Comparing between bge-reranker-v2-m3 and Qwen3-Reranker-4B with the baseline, we can see that the reranker mainly improves the top of the list. R@1 and R@5 go up by about 0.03, but R@10 barely moves. This means for the few questions the reranker improved, the answer was already in RRF's (Cormack et al., 2009) top 10, and the reranker moved it up. From the result, we can see that bge-reranker and Qwen-Reranker-4B has very similar performance. However, the gains are about 4 questions and not significant (p = 0.388/0.388 at k=1, 0.125/0.289 at k=5).

Reranker	R@1	R@5	R@10	p_1	p_5	p_{10}
None (RRF only)	0.702	0.810	0.843	–	–	–
BAAI/bge-reranker-v2-m3	**0.736**	**0.843**	0.851	0.388	0.125	1.000
Qwen/Qwen3-Reranker-0.6B	0.702	0.818	0.843	1.000	1.000	1.000
Qwen/Qwen3-Reranker-4B	**0.736**	**0.843**	**0.860**	0.388	0.289	0.625

Table 1: Comparison across different rerankers

	Dev, all (157)		Dev, long refs (36)		Leaderboard				
Style	F1	Recall	F1	Recall	Total	F1	Recall	ROUGE	LLM
short	0.713	0.707	0.304	0.229	53.33	51.85	47.15	43.67	3.83
routed	0.723	0.727	0.350	0.319	54.08	52.26	48.20	44.18	3.87
full	0.572	0.714	0.355	0.338	51.99	46.75	52.05	39.01	3.81
restate	0.335	0.749	0.475	0.497	65.18	56.51	79.58	51.29	3.93

Table 2: Comparison across different answer styles. The scale for Dev (public evaluation data) is 0 to 1, the scale for Leaderboard except LLM is 0 to 100, the scale for LLM is 1 to 5.

3.1.2 Answer Styles

In Table 2, I compared across different answer styles on both 157 public evaluation data and 150 leaderboard data. The motivation behind using other answer style rather than short comes from different answer distribution between public evaluation data and leaderboard data. We can see that on the dev data, the Recall of short is similar to the F1. However, on the leaderboard data the F1 is much higher than Recall. This implies that the leaderboard answer has longer length. This hypothsis is supported by the result. Recall on leaderboard data is short < routed < full (47.15 < 48.2 < 52.5). Also, Recall on dev, long reference data has large difference while overall dev recall has small difference. These observation combined tells that the answer on leaderboard is generally longer.

4 Analysis

4.1 Comparison on Difference Question Types

In Table 3 and Table 4, there are 4 types of major questions types: museum collection, event date, sports, and general. Table 3 is Recall @ 1 and Table 4 is Recall @ 5 result.

4.1.1 Museum Collection

These questions contain the artwork title and the short information of that artwork. Since my corpus include every artwork from the 2017 snapshot, this kind of questions are relatively simple. Every first-stage method achieves same performance (41/48) and only cross-encoder reranker improves by a little on classifying between near-duplicate titles ("The River" vs "By the River").

4.1.2 Event Dated

These questions are mostly who/what is on a given date. They depend on exact tokens, such as an event or people/group name with a specific date. This is why sparse method beats dense method at Recall @ 1. Sparse method looks for exact match, while dense embeddings may mixed up "November 5" and "November 21". However, at Recall @ 5, dense methods has better performance. This is because when sparse method finds the correct chunk, it is usually at rank 1 and when it misses, the chunk is far down. In contrast, dense methods tends to be close but not first. That is why dense will achieve better performance on Recall @ 5.

4.1.3 Sports

This type of question is easy for every methods. Most questions are just simple look-up of the player's name, so word matching method will have better performance. In Table 4, all methods except dense achieve 100% Recall@5.

4.1.4 General

These questions are the questions that don't fall in the categories above. Here the dense method has better performance because questions are phrased differently compared to the corpus, so semantic analysis did a better job. For example, one question is "How much does it cost to attend the Heinz History Center with a student ID?" and the corpus has "Students: $11." For sparse method, it is hard to match this question answer pair correctly.

Type	n	BM25	Dense	RRF	Weighted	Dense re-score	RRF + CE
museum_collection	48	0.854	0.854	0.854	0.833	0.854	**0.896**
event_dated	25	0.520	0.400	0.560	0.480	0.440	**0.640**
sports	18	0.778	0.722	**0.889**	0.833	0.722	0.833
general	20	0.300	0.500	0.450	0.400	0.400	**0.550**
All	121	0.661	0.636	0.702	0.653	0.628	**0.736**

Table 3: Recall @ 1 across different retrieval methods on various question types

Type	n	BM25	Dense	RRF	Weighted	Dense re-score	RRF + CE
museum_collection	48	0.875	0.875	0.875	**0.896**	0.875	**0.896**
event_dated	25	0.600	**0.720**	**0.720**	0.680	0.680	**0.720**
sports	18	**1.000**	0.944	**1.000**	**1.000**	**1.000**	**1.000**
general	20	0.600	0.650	0.600	0.600	0.550	**0.700**
All	121	0.785	0.793	0.810	0.818	0.777	**0.843**

Table 4: Recall @ 5 across different retrieval methods on various question types

RRF rank of answer	Questions
1	85
2–10	17
11–50	3
Not in top 50	16
Total	121

Table 5: RRF ranking of the correct answer of 121 short answer questions from public evaluation data

Step	ΔF1	Better / worse	p
Closed-book → RRF, top 5	+0.505	112 / 15	1.1×10^{-26}

Table 6: Comparison between closed-book, RAG, and rerankers

Retriever	R@1	R@5	R@10	p_1	p_5
BM25 (sparse)	0.661	0.785	0.818	–	–
Dense (BGE)	0.636	0.793	0.826	0.678	1.000
RRF	**0.702**	0.810	0.843	0.267	0.508
Weighted	0.653	**0.818**	**0.851**	1.000	0.125
Dense re-scoring	0.628	0.777	0.826	0.503	1.000

Table 7: Comparison of different retrieval methods on 121 short answer public evaluation data

4.2 System Variants

4.2.1 Reranker

In Table 1, it shows that the gain of reranker is small and not significant. This comes from small improvement room. Table 5 shows that 85/121 answers are already correctly ranked 1, and only 20/121 can be improved. The small improvement space makes the reranker gain not significant.

4.2.2 Answer Styles

In Table 2, the routed style looks very similar to the short style. This is because the longer output route is rarely triggered. The route is designed for descriptive questions captured by regular expression. It triggers on 13 of 157 public evaluation data questions, and all 13 do have long references. This is because some descriptive questions phrased like factoid questions. For example, the question "When does THAT LIKE THIS perform in Pittsburgh in 2026?" has full sentence answer rather than just a date.

We can also observe from Table 2 that different styles (length) gives similar content by looking at the LLM judge score and Recall score. The LLM score raised by 2.8 percentage points and Recall raised by 32.43 percentage points. Recall is the fraction of the correct answer's tokens that also appear in the prediction. Larger recall means our restated answers is closer to the references also use. However, the LLM score did not move much, which means the content of answers did not change much.

4.3 Closed-book vs RAG

In Table 6, retrieval has improve significantly ($p < 0.05$). It improves 112 of 157 questions. If we look at the questions that closed-book methods gets wrong, we can see that closed-book answers are confidently wrong rather than refusals. For example, for the question "Who is credited for "Where the Pavement Ends" in the CMOA catalog?", closed-book system answers "Robert Indiana" rather than "Corneille" or "Not sure". This shows hallucinations. With RAG, most of these

Question	Correct Answer	RRF, top 5	RRF + reranker, top 5
What team will the Pittsburgh Penguins play against on November 5, 2026?	Boston Bruins	No game scheduled (0.00)	Boston Bruins (1.00)
What accession number does the Carnegie Museum of Art list for the painting "By the River"?	1996.44	1996.44 (1.00)	1996.44 (1.00)

Table 8: Output and F1 scores across different systems

Question	Correct Answer	short	restate
What creation date does the Carnegie Museum of Art record for "Winter Bouquet"?	1992	1992 (1.00)	The Carnegie Museum of Art records the creation date for "Winter Bouquet" as 1992 (0.15)
Which major company was founded in Pittsburgh in 1869?	The H.J. Heinz Company, known for its food products, was founded in Pittsburgh in 1869.	Westinghouse Air Brake Company (0.11)	Westinghouse Air Brake Company was founded in Pittsburgh in 1869 (0.58)

Table 9: Output and F1 score across different answer styles

problems are solved.

4.4 Comparison Across Different Retrieval Methods

In Table 7, the RRF has the best Recall@1 score, while weighted has slightly better performance on Recall@5 and Recall@10. Note that in this table, nothing is significant. The reason for this is that the retrievers differ by only a few questions, and 121 questions is too few to detect differences that small. Adding more data will help evaluate the system more confidently.

We can see that hybrid method like RRF and Weighted defeat dense or sparse method on every metrics. The reason for this is that dense and sparse methods have their own weakness and fusion lets one method recover the other's misses. RRF uses ranks to fuse sparse and dense results, while Weighted method use scores to fuse the two results, taking confidence into account. In other words, the RRF has better performance when both methods rank the correct chunk near the top. In contrast, the Weighted method wins when one list ranks the correct chunk highly, and the other does not. Despite understanding the underlying mechanism, the two are statistically indistinguishable on this data.

4.5 Examples of Outputs

4.5.1 Reranker

In Table 8, I report one question that reranker system actually has improvement on and another question that both systems responded correctly. In question "What team will the Pittsburgh Pen- guins play against on November 5, 2026?", the system without reranker cannot answer because the correct information is out of top 10 range. In contrast, with reranker, this chunk is moved to rank one.

4.5.2 Answer Styles

In Table 9, we can see there is one question that short style answer prevails and another question that short style loses. The content of them are all corect, it is just the format that cause large difference in F1 score.

5 Use of AI

I use Claude Code as my coding agent to help me to understand the homework, provide data source suggestions, write code, and run the code.

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